CVRP with Sequential Quantum Subset Sampling

DuQuantum / Pasqal | Track A | Local review draft | 25 October 2026

1. Grouping customers within vehicle capacity

We assign every customer to one vehicle, keep each load within capacity Q, and evaluate total depot-to-depot travel. Track A uses quantum-emulator measurements to propose customer groups. Routes within those groups are computed exactly by classical code for evaluation.

The original benchmark has six two-dimensional instances: S/M/L with seeds 11 and 29, customer positions in $[0,100]^2$, and depot (50,50). A checked cyclic assignment certifies feasibility but is not given to the sampler. The saved results use equal demands; later heterogeneous-demand tuning is reported separately.

Tier	Customers N	Vehicles K	Capacity Q	Demand	Load ratio
S	4	2	7	3	0.8571
M	10	3	20	5	0.8333
L	15	4	30	6	0.7500

Clustering score. S denotes a customer group; |S| is its size. The normalized distance $w_{ij}=d_{ij}/(100\sqrt{2})$ lies in [0,1]. The score rewards compact, balanced groups. It does not include depot travel or visiting order:

$$F = \sum_{\text{group } S} \left[0.15|S|^2 + \sum_{\substack{i<j \\ i,j\in S}} w_{ij} \right]$$

Assignment and unused capacity. $x_{ik}=1$ assigns customer i to vehicle k. Each customer needs exactly one vehicle. Slack s_k is unused vehicle capacity; d_i is demand. Coverage and capacity require:

$$A\left(\sum_k x_{ik} - 1\right)^2 \qquad \sum_i d_i x_{ik} + s_k = Q, \quad s_k \geq 0$$

$$s_k = \sum_\ell 2^\ell u_{k\ell} \qquad s_k = \sum_{\ell=0}^{B_Q-1} 2^\ell u_{k\ell}, \qquad B_Q = \lceil \log_2(Q+1) \rceil$$

Binary $u_{k,l}$ encodes slack using $B_Q=\text{ceil}(\log_2(Q+1))$ bits. Adding squared coverage and capacity residuals to F gives an exact target QUBO:

$$E_{\text{slack}} = F(x) + A\sum_i \left(\sum_k x_{ik} - 1\right)^2 + B\sum_k \left(\sum_i d_i x_{ik} + s_k - Q\right)^2$$

For nonnegative F and a checked feasible score U, A=B=U+1 is sufficient for all exact target global minima to be feasible. This reference construction is not the executed capacity Hamiltonian. Our sampler avoids slack atoms by changing the subproblem and rejecting invalid measured choices.

2. Sequential sampling and physical implementation

At each vehicle stage, R is the unassigned customer set and L the remaining fleet. A seeded window contains at most eight customers; b_i=1 selects a customer and m counts the selected set. The baseline uses lambda=0.15 and energy scale kappa=4:

$$E_{\mathrm{cl}}/\kappa = \lambda m^2 - \mu m + \sum_{i<j} w_{ij} b_i b_j \qquad m = \sum_i b_i, \qquad \mu = 2\lambda \frac{|R|}{L}$$

The size-only term is minimized near m=|R|/L. Expansion gives shared h_i=kappa(lambda-mu) and positive J_{ij}=kappa(2lambda+w_{ij}). This favors plausible subsets, not guaranteed-capacity solutions. Accepted customers are removed; the final vehicle receives the remainder if it fits. There is no repair or backtracking.

Binary-variable count	S	M	L
Direct joint N^2K	32	300	900
Assignment NK	8	30	60
Assignment + slack	14	45	80
Executed clustering peak	4	8	8

Atom energy. n_i counts Rydberg excitation and X_i mixes the two states. The laser drive Omega prepares the state; detuning delta sets linear costs, and atom distances set repulsive pair costs:

$$\frac{H(t)}{\hbar} = \frac{\Omega(t)}{2} \sum_i X_i - \sum_i \delta_i(t) n_i + \sum_{i<j} \frac{\Gamma}{r_{ij}^6} n_i n_j$$

$$h_i^{\mathrm{real}} = -\delta_i(T), \qquad J_{ij}^{\mathrm{real}} = \frac{\Gamma}{r_{ij}^6}$$

Track A uses one global detuning for all atoms. At Omega(T)=0, positive detuning lowers the cost of excitation; interactions charge for selecting pairs. Gamma=C_6/hbar is about 865723.02 rad/us micrometer^6. SLSQP fits the physical couplings, with at most 120 iterations and a legal-grid fallback. Separation >=5 micrometers and radius <=38; all long-range interactions remain.

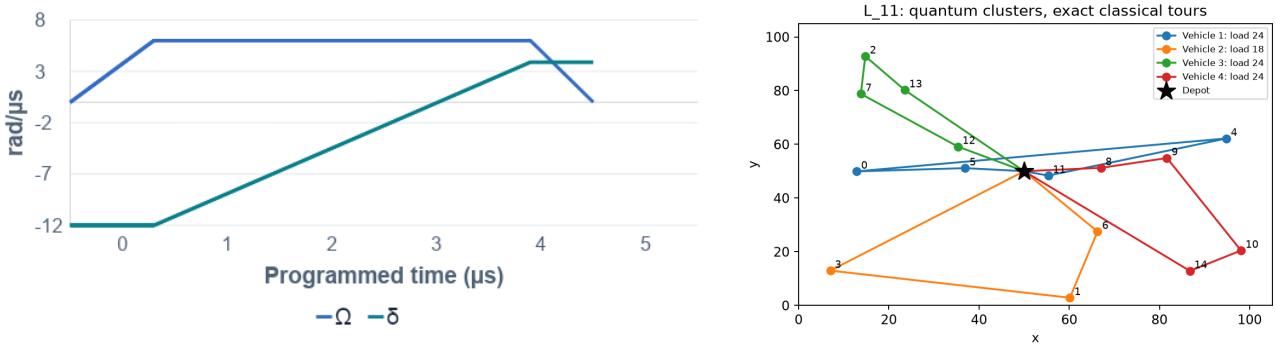


Figure 1. Left: programmed slow pulse, 5.2 microseconds. Right: saved L_11 groups with exact classical tours; colors identify vehicles, and the star is the depot. The customer map and physical atom layout are different geometries.

Baseline $\mathrm{Omega}_{\mathrm{max}}$=6 rad/us and initial delta=-12. Fast/slow rise-sweep-fall schedules are 400/1200/400 and 800/3600/800 ns, within the 6000 ns envelope. Pulser validates controls. QutipBackendV2 runs noiseless CPU state-vector emulation; DMM, modulation response and calibrated layouts are not used here. Relative coupling error is $||J^{\mathrm{real}}-J^{\mathrm{target}}||_2/||J^{\mathrm{target}}||_2$.

3. Baseline results: feasibility and solution quality

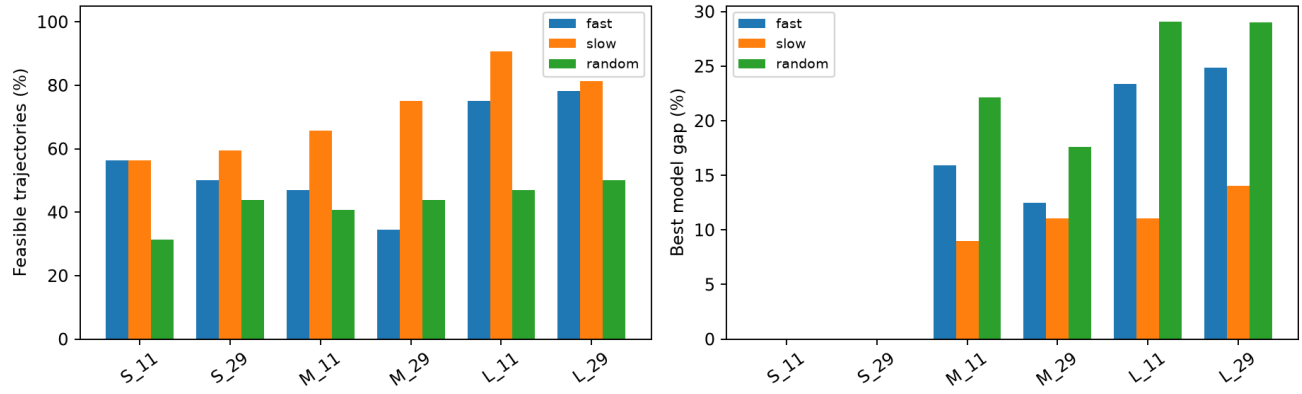


Figure 2. Original six-instance benchmark. Left: valid complete assignments, higher is better. Right: best clustering-model gap, lower is better. These bars compare fast/slow pulses with random-bit proposals, not equal total compute budgets.

Instance	Method	Feasible %	Model gap %	CVRP gap %
S_11	fast	56.2	0.00	0.00
S_11	slow	56.2	0.00	0.00
S_11	random	31.2	0.00	0.00
S_29	fast	50.0	0.00	0.00
S_29	slow	59.4	0.00	0.00
S_29	random	43.8	0.00	0.00
M_11	fast	46.9	15.91	20.52
M_11	slow	65.6	9.00	21.95
M_11	random	40.6	22.12	15.27
M_29	fast	34.4	12.47	15.09
M_29	slow	75.0	11.07	16.12
M_29	random	43.8	17.62	22.44
L_11	fast	75.0	23.36	43.89
L_11	slow	90.6	11.05	27.41
L_11	random	46.9	29.06	41.89
L_29	fast	78.1	24.85	35.99
L_29	slow	81.2	14.05	28.49
L_29	random	50.0	29.02	37.69

Each method uses 32 complete trajectories resampled from cached 1000-shot conditional banks. Failures stay in the feasibility/reference-hit denominators. Model gap compares F with the exact constrained F optimum; CVRP gap compares the best exact-routed sampled grouping with the full CVRP optimum. The two columns may select different trajectories.

$$\mathrm{gap}(\%) = 100\left(\frac{C}{C^*} - 1\right)$$

Use the formula at the named objective level; no feasible output means an undefined gap, not zero. Reference-hit tolerance is 0.000001 percent. References enumerate feasible subsets and solve exact partition DP; S also checks all assignments and CPLEX (30 s, one thread, zero relative MIP gap). Vehicle relabellings are removed. In evaluate, exact routed cost breaks ties between equal F scores; it does not repair a measured choice.

4. Tuning, interpretation, and what did not work

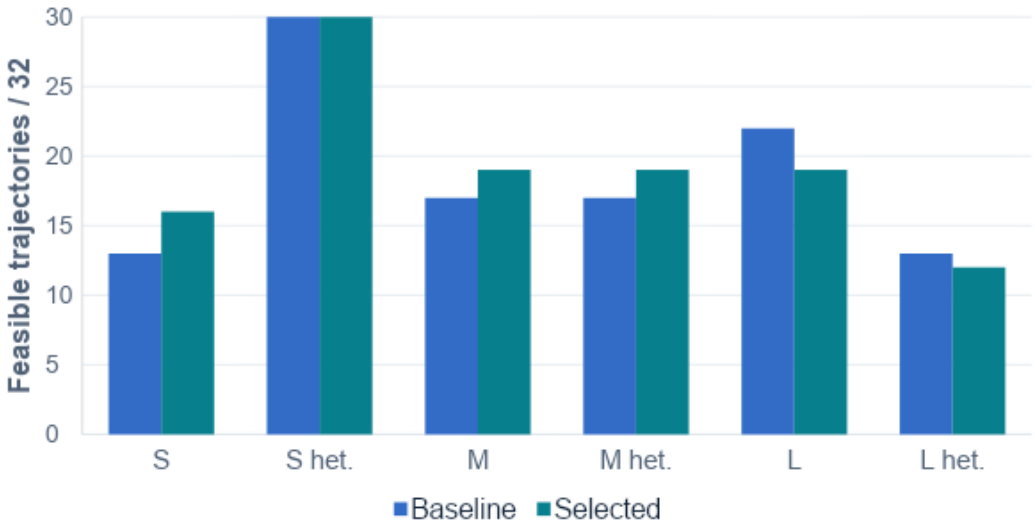


Figure 3. Held-out Track A tuning experiment. Bars count valid trajectories out of 32 for the baseline and selected settings. S/M/L and their heterogeneous-demand variants use seed 101, separate from training seed 11.

Six held-out cases	Baseline	Selected
Mean feasible trajectories	58.85%	60.42%
Mean best model gap	8.41%	7.57%
Total solver time	18.59 s	43.42 s

Sixteen settings were screened before full training and held-out evaluation. The selected setting uses Omega=8, scale=2, reward multiplier 0.8, a 5200 ns pulse and three embedding starts. Its average grouping score improves, but runtime rises and not every instance improves. Geometry fitting and different backend-call counts make this a joint tuning experiment, not an equal-compute ablation.

Energy diagnostic. The separate identical-QUBO analysis found a sampled target optimum in 707/754 clustering banks; none of the 754 fitted native models retained a target ground-state bitstring. A low-energy physical state is therefore not automatically the intended best grouping. Finite-time evolution can still produce useful measured states.

What did not work. Log-coupling fitting produced large errors and zero large-tier feasibility in the exploratory run; squared-coupling fitting improved the recorded baseline. Planar embedding still distorts the objective. The surrogate ignores depot tours, while sequential choices and fixed windows limit search. On M_11, slow pulses improve F over random but give worse travel distance. A slower pulse cannot fix the wrong model.

The current heterogeneous-capable checks reject selected overloads and insufficient remaining total capacity; that necessary bound does not solve bin packing. Cached banks share estimation uncertainty, and two location seeds do not support a speedup claim. Next work is better model/embedding fit and evaluation across more independent seeds.

Interactive demonstration. Atlas Orbit visualizes the same measured grouping decisions and capacity checks before Track B orders the stops. Its optimum-reaching 3D result is a separate, higher-budget experiment, described and pictured in the Track B report. Demo source and replay instructions.

AI assisted formulation, coding, debugging and writing.

Sources: Pasqal challenge v3.0 (24 Oct 2026); team presentation (25 Oct 2026); challenge/results/results.csv, MODEL.md and TUNING.md; Lucas (2014), doi:10.3389/fphy.2014.00005; Pulser 1.9.1. Reproduction commands and pinned dependencies: README.md.